# COOL-SEASON OROGRAPHIC SNOWFALL EXTREMES IN THE CENTRAL WASATCH MOUNTAINS, UTAH, USA

by

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ABSTRACT

Heavy orographic snowfall can disrupt transportation and threaten life and property in mountainous regions but also provides benefits for water resources, winter sports, and tourism. Frequently experiencing orographic snowfall extremes, Little Cottonwood Canyon (LCC) in the central Wasatch mountains of northern Utah is one of the snowiest locations in the interior western United States. Due to a high density of avalanche paths, heavy snowfall, and extensive vehicle traffic, state route 210 (SR-210) in LCC has the highest uncontrolled avalanche hazard index of any major road in the world, with frequent closures and threats to public safety.

Using manual 12-h snow and liquid precipitation equivalent (LPE) observations, ERA5 reanalyses, and operational radar data, this thesis examines the characteristics of cool-season snowfall extremes in upper LCC. Nearly all cool-season precipitation in upper LCC falls as snow and 12-h extremes, defined based on new snow or LPE, occur for a wide range of crest-level flow directions, with the former most frequent during west-northwesterly flow and the latter most frequent during south-southwesterly flow. Both are produced by several synoptic patterns, including high integrated vapor transport ($IVT$), frontal systems, post-cold-frontal northwesterly flow, and closed low pressure systems. Post-cold-frontal northwesterly flow events efficiently convert $IVT$ to precipitation and can be highly localized with radar echoes confined to the central Wasatch. Depending on the atmospheric environment, high $IVT$ can either produce no

LPE or heavy LPE. These results illustrate the remarkable diversity of mechanisms for orographic snowfall extremes in LCC.

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# 1. INTRODUCTION

Orographic precipitation extremes in the form of heavy rainfall, snowfall, or other forms of solid precipitation can produce wide-ranging geophysical hazards and societal impacts including flooding (Maddox et al. 1978; Ralph et al. 2006; Neiman et al. 2011; Froidevaux and Martius 2016), avalanches (Schweizer et al. 2003, 2009; Farestveit and Skutlaberg 2009; Conlan and Jamieson 2016a,b), landslides (Johnston et al. 2021), structural collapses (Hilker et al. 2009), power outages (Roebber and Gyakum 2003), and immobilized transportation (Nöthiger and Elsasser 2004). Such extremes occur frequently during moist flow over hills and mountains with orographic lift enhancing precipitation rates through processes that include seeder-feeder and potential instability (Roe 2005; Houze 2012; Colle et al. 2013; Stoelinga et al. 2013). Orographic precipitation extremes have been tied to atmospheric rivers (Neiman et al. 2011; Lavers and Villarini 2013; Froidevaux and Martius 2016; Lorente-Plazas et al. 2018), convective storms (Caracena et al. 1979; Wulfmeyer et al. 2011; Casaretto et al. 2022), lake and sea-effect precipitation (Manabe 1957; Campbell et al. 2018; Veals et al. 2020; Steenburgh and Nakai 2020), extratropical cyclones (Lackmann and Gyakum 1999; Pandey et al. 1999; Viale and Nuñez 2011), and tropical cyclones (Smith et al. 2009; DeHart and Houze 2017), including those undergoing extratropical transition (Sinclair 1993; Sturdevant-Rees et al. 2001; Stohl et al. 2008; Liu and Smith 2016; Lentink et al. 2018). Often, the intensity and spatial distribution of orographic precipitation is closely related to wind direction, with precipitation heaviest along or upstream of ridgelines that are aligned orthogonal to the flow (Houze et al. 2001; James and Houze 2005; Yuter et al. 2011; Foresti et al. 2018). For many regions — such as North America's Cascades, Olympics, and Sierra Nevada or the European Alps — distinct synoptic patterns directly influence the location of extreme orographic precipitation (e.g., Pandey et al. 1999; Minder et al. 2008; Yuter et al. 2011; Froidevaux and Martius 2016); in contrast, *several* different synoptic types can produce orographic precipitation extremes in some regions like Italy's Apennine Mountains (e.g., Capozzi et al. 2022).

Cool-season orographic precipitation extremes occur frequently in the Wasatch mountains of northern Utah, a narrow, quasi-linear, and quasi-meridionally oriented mountain range at the eastern periphery of the Great Basin in the interior western United States (Fig. 1). The Wasatch Range broadens east of the Salt Lake Valley (SLV) where a series of zonally oriented ridges (Alpine, Cottonwood, and Wildcat) and canyons [Little Cottonwood (LCC), Big Cottonwood (BCC), and Mill Creek (MCC)] compose a significant portion of what is known locally as the central Wasatch. Little Cottonwood Canyon splits the Alpine and Cottonwood ridges, which rise abruptly to ~3300 m MSL, more than 2000 m above the SLV and Great Salt Lake (GSL). Mean cool-season (Oct–Apr) liquid precipitation equivalent (LPE) and snow increases from 432 mm and 207 cm at Cottonwood Weir (CW; see Fig. 1; 1520 m MSL) at the western base of the central Wasatch to 902 mm and 1106 cm at Alta (2655 m MSL) in upper Little Cottonwood Canyon (NCEI 2021).

Within LCC, a two-lane highway, State Route 210 (SR-210), is the only transportation link to Alta. The road climbs 1000 m in 13 km, crossing 50 avalanche paths that leave it vulnerable to avalanches during frequent cool-season storms (Steenburgh 2023). Due to heavy traffic volumes, a lack of control structures, and the large concentration of avalanche paths, SR-210 has the highest uncontrolled avalanche hazard index (Schaerer 1989) of any major road in the world (Nalli and Mckee 2018). During and after periods of heavy snowfall, dangerous avalanche conditions can force the closure of the SR-210 and induce interlodge conditions during which residents and visitors to Alta and the nearby Village of Snowbird are legally mandated to remain indoors (Steenburgh 2023); during a particularly intense storm in February 2020, interlodge lasted for 52 consecutive hours. During the 2022/23 cool season, extreme snowfall and avalanches forced frequent multiday closures of SR-210, stranding residents and visitors and leading to food shortages (Jag 2023). The estimated revenue loss for ski areas and other businesses in upper LCC during the closure of SR-210 was \$1.4 million day$^{-1}$ in the 1991/92 season (Blattenberger and Fowles 1995), or about \$3 million day$^{-1}$ in 2023 dollars.

Prior work shows that heavy precipitation and snow in the Wasatch can be associated with 1) unstable, post-cold-frontal northwesterly flow, 2) cold-frontal passages, 3) decaying or inland penetrating atmospheric rivers 4) localized lake-effect convection, and 5) closed upper-level low pressure systems (Dunn 1983; Carpenter 1993; Horel and Gibson 1994; Steenburgh 2003; Rutz and Steenburgh 2012; Alcott et al. 2012; Rutz et al. 2015). Dunn (1983) identified a maximum in the frequency of high LPE snowfall events at Alta with west-northwest flow (~290˚) and speculated that this was due to low level convergence and lake-effect convection. Northern Utah features a local maximum in strong cold-frontal passage frequency in the Intermountain West, as delineated by an abrupt temperature fall and pressure rise (Shafer and Steenburgh 2008). In some Wasatch storms, cold-frontal passages can feature an intrusion of low-$\theta_e$ air aloft, enabling heavy convective precipitation in the pre-frontal phase (Steenburgh 2003). This has also been observed in the Tushar Range of southern Utah (Long et al. 1990; Sassen et al. 1990). Cold-frontal passages can also be accompanied by heavy precipitation and be followed by post-frontal orographic and GSL-effect precipitation (Steenburgh 2003). Though it only represents about 5% of the total cool-season precipitation in the central Wasatch, the GSL-effect can produce heavy snow, especially in the early and latter parts of the cool season (Carpenter 1993; Steenburgh et al. 2000; Alcott et al. 2012; Alcott and Steenburgh 2013; Yeager et al. 2013; McMillen and Steenburgh 2015a,b). Prior research has linked inland penetrating or decaying atmospheric rivers from the Pacific Ocean with orographic precipitation in the Wasatch, although they have not been examined systematically (Dettinger et al. 2011; Rutz and Steenburgh 2012; Rutz et al. 2014, 2015). Intermountain cyclones have also been tied to precipitation in the Wasatch (Zishka and Smith 1980; Whittaker and Horn 1981; Dunn 1983; Horel and Gibson 1994), often causing a decreased ratio of mountain to valley precipitation (Williams and Peck 1962; Dunn 1983).

Previous research has examined individual phenomena which contribute to extreme orographic snowfall in the Wasatch, but no studies have comprehensively analyzed orographic snowfall characteristics and synoptic regimes in the range — in this thesis, we aim to fill this gap. Additionally, most prior research on orographic precipitation only considers meteorological patterns that produce heavy precipitation in the form of liquid precipitation equivalent, and as we will show here, there is considerable variation in the meteorology associated with extreme snow compared with heavy LPE in the central Wasatch. The purpose of this research is to conduct a thorough study of orographic snowfall in the central Wasatch and LCC. Our results will aid forecasting, avalanche hazard assessment, and road-weather maintenance in the central Wasatch and enhance knowledge of the mechanisms contributing to orographic precipitation extremes in a narrow, interior mountain range. In Section 2, we describe the data and methods used. In Section 3, we present the event and seasonal characteristics of snowfall in upper LCC and discuss the flow dependencies, vapor-transport tendencies, and synoptic environments enabling extreme precipitation in the central Wasatch. A radar analysis presents the local to mesoscale characteristics of extreme precipitation periods. Section 4 summarizes our results.

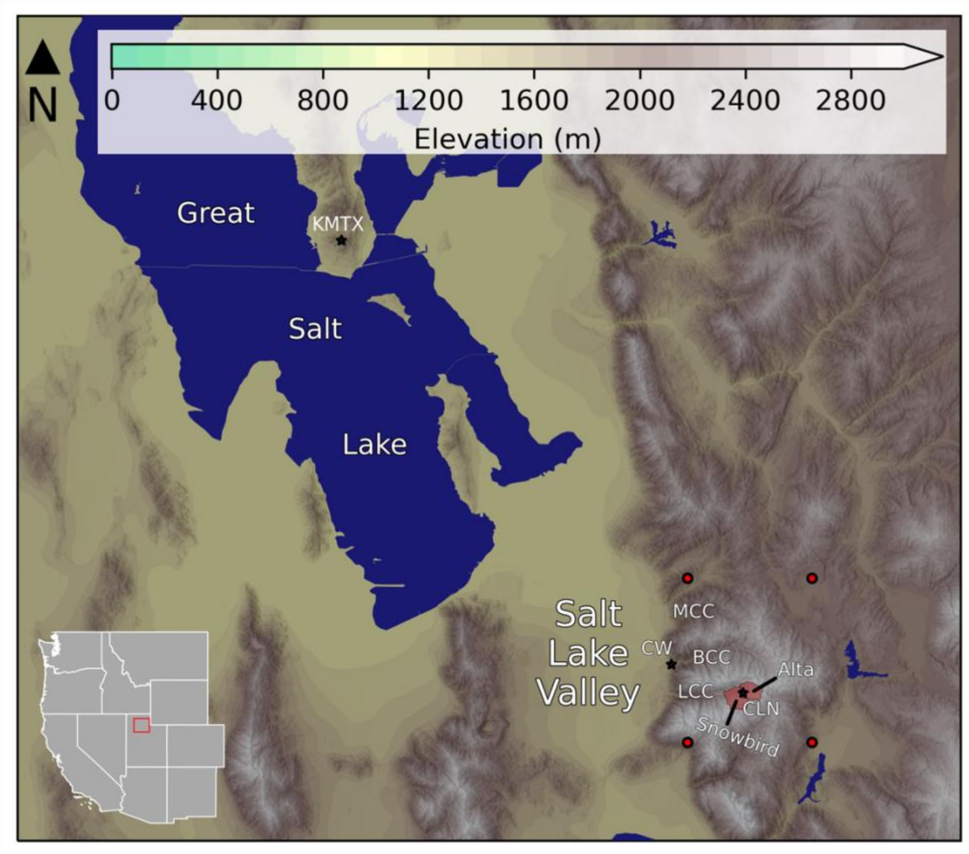


Figure 1. Topography of the central Wasatch and surrounding areas (scale at top) with key geographic features annotated and four ERA5 grid points used to determine meteorological characteristics identified by red dots. The location of the map within the western United States is shown at bottom left.

# 2. DATA AND METHODS

## Alta-Collins (CLN) Snowfall Observations

We used cool-season (Oct–Apr) manual new snow and LPE observations collected by the Alta Ski Patrol at their Alta-Collins (CLN) snow-study plot (2945 m MSL; see Fig. 1 for location) from 1999–2022, consisting of 23 seasons of data. This dataset contains 12-h observations of new snow and LPE taken daily at 0400 and 1600 LST. Twenty-four-hour new snow, LPE, and resulting snow-to-liquid ratio observations from this site, which are collected on a standard snowboard placed on the existing snowpack, were used previously by Alcott and Steenburgh (2010) for a study of snow-to-liquid ratio at Alta, but we use 12-h observations here since they are now being provided by Alta Ski Patrol and there is less evolution in the synoptic conditions during the shorter period. Sampling methods and the fact that nearly all of the cool-season precipitation at this altitude falls as snow in the present climate ensure that the LPE observations represent the water-equivalent of snowfall. We quality controlled the data through comparison with precipitation and snowfall observations at the nearby Alta National Weather Service COOP site (2655 m MSL) and the Utah Department of Transportation Alta Guard Station (2682 m MSL) and by examining synoptic reanalyses during event periods. There were three measurements that we deemed questionable, and we either adjusted the date and time to match nearby observations and reanalysis data or eliminated the observation entirely (Table 1). Each questionable event occurred in October, before

Alta Ski Area opened for the season, which may have contributed to a mislabeling of an observation date or measurement. We defined 12-h periods with ≥ 2.54 cm of new snow as a "snow event" and ≥ 2.54 mm of LPE as an "LPE event." We identified the 95th percentile of new snow or LPE during these events and classified them as orographic snowfall extremes. Thus, these are 12-h snowfall extremes rather than 24-h or multiday events.

ERA5 Reanalysis

To classify the meteorological conditions associated with orographic snowfall extremes at CLN, we used the European Centre for Medium-Range Weather Forecasts Reanalysis v5 (ERA5; Hersbach et al. 2020). The ERA5 contains global meteorological variables on a 0.25° grid at hourly intervals on pressure levels ranging from 1000 hPa to 1 hPa. There are four ERA5 grid points that surround the central Wasatch Range (see Fig. 1 for locations), and we averaged data over those four grid points to classify the local meteorological conditions during events at CLN at the mid-point of a 12-h observation period. Sensitivity testing revealed that using other times during the period or time averages did not substantially impact results. We binned all events into 1 of 16 wind directions using the 700-hPa (the closest mandatory pressure level to crest level) wind direction. We also manually classified all 95th percentile snow and LPE events into 1 of 7 synoptic categories by examining synoptic-scale analyses of 500-hPa geopotential height and absolute vorticity, sea level pressure, 3-h accumulated precipitation, 700-hPa temperature, 700-hPa relative humidity, 700-hPa wind, and vertically integrated vapor transport ($IVT$) magnitude and vectors at the mid-point of each event. We calculated $IVT$ from the ERA5 surface to 300-hPa using the equation

$$IVT = \sqrt{\left(\frac{1}{g}\int_{sfc}^{300} qu\,dp\right)^2 + \left(\frac{1}{g}\int_{sfc}^{300} qv\,dp\right)^2}, \tag{1}$$

where $q$ is specific humidity in kg kg$^{-1}$, $u$ and $v$ are the zonal and meridional wind in m s$^{-1}$, $p$ is pressure in Pa, and $g$ is the acceleration due to gravity.

## KMTX Radar

To examine the characteristics of mesoscale precipitation features during orographic snowfall extremes at CLN, we used level II data (NOAA 1991) from the Salt Lake City (KMTX) WSR-88D radar located at 2000 m MSL on Promontory Point above the Great Salt Lake (Fig. 1a). Because the resolution of the radar data increased to super-resolution before the 2008–2009 cool season, radar statistics are based on events beginning with that season. For extreme (95$^{th}$ percentile) LPE events, none of the 75 events that occurred during and after the 2008/09 cool season had missing radar data, and for extreme snow events, data were missing for only one of 83 events (1.2%). To study the spatial distribution of precipitation during cool-season storms in the central Wasatch and upstream valleys and ranges, we examined the frequency of radar echoes ≥ 10 dBZ for all available 0.5° radar scans during events. We used a 10 dBZ threshold because it represents a threshold for accumulating snow and has been used in prior studies in northern Utah (e.g., Steenburgh et al. 2000; Yeager et al. 2013). As a mountaintop radar in a mountainous region, KMTX often experiences beam blockage due to the Wasatch mountains, and radar scans are occasionally too high to scan shallow storms, potentially limiting our results (Wood et al. 2003). Thus, we use a fixed dBZ threshold, rather than an LPE estimate using a $Z - S$ relationship, enabling us to characterize the spatial mesoscale precipitation *patterns* associated with snowfall extremes, rather than quantifying them using the radar data.

Table 1. Adjustments made to individual events in the CLN snow and LPE observations.

| **Original Event End Time (LST)** | **New Event End Time (LST)** |
|---|---|
| 17 Oct 2004 0400 | Removed from analysis |
| 30 Oct 2004 1600 | 31 Oct 2004 1600 |
| 27 Oct 2010 0400 | 27 Oct 2010 1600 |

# 3. RESULTS

Event and Seasonal Snowfall Characteristics

We begin with an examination of event and seasonal snow and LPE characteristics at CLN in upper LCC. We identified 2707 snow events over the 23-year period, with a mean annual number of events of 117.7 (Fig. 2a). The mean and median 12-h snowfalls were 11.2 cm and 7.6 cm, respectively, with over 1700 events containing < 12.5 cm of snow and only 10 containing ≥ 48.5 cm of snow, the largest of which had 64.8 cm. There were 138 $95^{th}$ percentile events, each with ≥ 30.5 cm of snow over a 12-h period. Our dataset contains 2304 (a mean of 100.2 per season) LPE events (Fig. 2b). Unsurprisingly, the structure of Fig. 2b is similar to Fig. 2a, with an abundance of small events, and only 10 that exceeded 48.5 mm of LPE over a 12-h period, with a maximum of 70.6 mm. The mean event LPE was 10.6 mm, and the median was 7.9 mm. There were 116 $95^{th}$ percentile events, each with ≥ 27.9 mm of LPE over a 12-h period. Only 52 events were both $95^{th}$ percentile snow *and* LPE events.

There is a lack of variation in sub-seasonal snow and LPE at CLN, as most months tend to observe similar snow and LPE amounts (Fig. 3). For both snow and LPE, each month from December to April observes median snow and LPE values that are not statistically different at the $95^{th}$ percent confidence interval (Fig. 3). Median monthly snow from December to April is roughly 200 cm, and median LPE during those same months is around 175 mm. October tends to mark the transitional period to a more cool-season pattern in the central Wasatch; in some years, it can see substantial snow and LPE, while it is also possible to see no precipitation during that month.

Featuring an intermountain snow climate, the central Wasatch observes high seasonal snowfalls and moderate wintertime temperatures (Whiteman 2000; Mock and Birkeland 2000). In select seasons, it experiences elements of continental or coastal snow climates (Mock and Birkeland 2000). During the 23-season Oct–Apr study period, snow at CLN ranged from 831.9 cm to 1877.1 cm and LPE ranged from 726.4 mm to 1681.7 mm (Fig. 4). The mean seasonal snow was 1340.8 cm and the mean seasonal LPE was 1116.9 mm (Fig. 4). The 2011 cool season featured the most events, with 181 snow events and 196 LPE events, while the 2015 cool season, a drier period, only had 96 snow events and 109 LPE events (Fig. 4).

Indicative of the large variety of storm tracks and types that produce orographic snowfall extremes in the central Wasatch, there is a wide range of snow-to-liquid ratios (SLR) in individual storms at CLN. We generated SLR statistics for events with ≥ 5.1 cm of snow and ≥ 2.8 mm of LPE, and there were 1903 such events in our 23-season study period. These criteria reduce relative errors in SLR due to rounding and measurement (Judson and Doesken 2000; Roebber et al. 2003; Baxter et al. 2005). Snow-to-liquid ratio in these events ranged from 1.6 to 50, with a mean of 13.8 and a median of 12.8 (Fig. 5), comparable to results from Alcott and Steenburgh (2013), who found a mean of 14.4 and a median of 13.3 in similar analysis of SLR at CLN for Nov–Apr 1999–2007. Our lower values are likely due to our inclusion of October in our analysis and the use of observations through 2022.

The El Niño Southern Oscillation (ENSO) is an important driver of weather and climate variability in the continental western United States (Bjerknes 1966, 1969; Rowntree 1972; Horel and Wallace 1981; Ropelewski and Halpert 1986; Trenberth et al. 1998). While ENSO phase can have seasonal precipitation impacts on regions in the American west [e.g., positive ENSO has links to increased precipitation in California (Cayan et al. 1999; Jong et al. 2016), and negative ENSO has links to increased precipitation in the Pacific Northwest (Redmond and Koch 1991)], northern Utah's mountains fall at the nodal point in the ENSO dipole, where ENSO phase has little impact on seasonal precipitation trends (Redmond and Koch 1991). Using snow and LPE data at CLN and the multivariate ENSO index version 2 (MEI.v2; available at https://psl.noaa.gov/enso/mei; Wolter and Timlin 2011), we analyzed the relationship between ENSO index and two-month accumulated snow (Fig. 6a) and LPE (Fig. 6b). Consistent with prior research, ENSO has no discernable impact on observed snow or LPE at CLN. In any two-month sub-seasonal period, strong positive ENSO periods (i.e., ENSO index ≥ 1.5) and strong negative ENSO periods (i.e., ENSO index ≤ -1.5) do not produce any more snow or LPE at CLN than neutral phases or weak periods (Fig. 6). The Pearson correlation coefficients between two-month ENSO index and snow and LPE index were $r_{xy} = -0.03$ and $r_{xy} = -0.01$, respectively, indicative of the lack of relationship between ENSO and snowfall in the central Wasatch.

Flow Dependencies

We now present the characteristics of crest-level flow and associated temperatures and SLR which contribute to orographic precipitation extremes in the central Wasatch.

Cool-season climatological wind directions at 700 hPa (near crest level) in the central Wasatch most frequently range from southerly to northwesterly, with wind speeds ranging from 5-10 m $s^{-1}$ (Fig. 7a). There is very weak bimodality in the climatological wind direction, with maxima in northwesterly and south-southwesterly flow. When considering the frequency of flow directions for *all* snow and LPE events, there are also two maxima in west-northwesterly and south-southwesterly flow (Fig. 7b,c), roughly aligning with the cool-season climatology (Fig. 7a), although the bimodality strengthens compared to the climatology. While extreme (95$^{th}$ percentile) snow events occur across a wide range of 700-hPa flow directions, there is clear bimodality in the flow directions with greatest frequency, with the primary maximum in west-northwesterly and northwesterly flow, likely due to post-frontal orographic forcing mechanisms or lake-effect precipitation, which produce snow with relatively high snow-to-liquid ratios (Figs. 7d; 8a,b). In contrast, extreme LPE events are most common in southerly or south-southwesterly flow, although these events can also occur across a wide range of flow directions (Fig. 7e). With warmer temperatures, southwesterly flow orographic snowfall extremes typically to have lower SLRs, producing exceptional LPE but snowfall that would not be a 95$^{th}$ percentile snowfall event (Figs. 8; 9). These results contrast observations from other mountainous regions throughout the world, where there is limited variability in flow directions that produce orographic precipitation extremes (e.g., Pandey et al. 1999; Yuter et al. 2011; Foresti et al. 2018), but are consistent with other regions (e.g., Capozzi et al. 2022). Additionally, the wind speeds of the two maxima in extremes are distinct, as the west-northwesterly one occurs with wind speeds 5-15 m $s^{-1}$, and the south-southwesterly one has higher wind speeds of 10-20 m $s^{-1}$ (Fig. 7d,e).

To examine the representativeness of the mid-point for event classification (see Section 2), we next examine the magnitude of wind direction variability during extreme orographic snowstorms at CLN. Using the mid-point of each event, we identified LPE and snow events with flow directions ranging from southeasterly to north-northwesterly. Because the large majority of events were classified as either SSW, SW, WSW, W, WNW, or NW, we decided to focus subsequent analysis on events with those flow directions. Most events have little wind direction variability, since the wind bin with the highest normalized frequency corresponds with the wind direction subfigure for each flow direction in Figs. 10 and 11. The 700-hPa wind direction during events has a tendency to veer with time, with veering occurring more frequently during SSW, SW, WSW, and W events, which tend to feature a trough passage and a distinct wind shift (see Fig. 10a-d and Fig. 11a-d). On the contrary, events with WNW and NW flow have greater flow persistence during the course of an event (see Fig 10e-f and Fig. 11e-f), largely due to orographic snowfall extremes with these flow directions being associated with post-frontal orographic forcing.

Synoptic Classifications of Extreme Events

To examine the synoptic environments generating extreme orographic snowfall in the central Wasatch, we manually classified extreme snow and LPE events at CLN in one of seven synoptic types, using the nine variables in the ERA5 reanalysis listed in Section 2. The seven synoptic types were high $IVT$ with southerly (SIVT), southwesterly (SWIVT), westerly (WIVT), or northwesterly (NWIVT) flow, post-cold-frontal northwesterly flow (NWPF), frontal events (FR), and closed low pressure systems (CL).

For snow and LPE, less than 10% of events were unclassifiable and did not fall clearly into one of the above synoptic types (Fig. 12). Extreme LPE events tended to fall into one of the four high vapor transport categories more frequently than extreme snow events (Fig. 12), likely because high $IVT$ events have lower SLR values. Nearly 50% of snow events are associated with post-cold-frontal northwesterly flow, while only 16% of LPE events were NWPF (Fig. 12). The remaining events were FR or CL, representing about 15% of LPE and snow events (Fig. 12).

We produced composites of $IVT$, 500-hPa geopotential height, and 700-hPa temperature and winds for each synoptic type for both snow and LPE extremes. The appearance of the composite of each synoptic type was quite similar regardless of if we analyzed LPE or snow events. For brevity, we show only the seven composites for 95$^{th}$ percentile LPE events. For each high $IVT$ synoptic type, the location of the highest $IVT$ penetration in the interior is either north or south of the high Sierra Nevada where there is an $IVT$ minimum in the lee (Fig. 13a-d). For the SIVT and SWIVT composites, high $IVT$ values circumscribe the southern High Sierra and penetrate into northern Utah via the lower Colorado River Basin (Fig. 13a-b). For the WIVT composite, high $IVT$ values extend across the northern Sierra Nevada and southern Cascades to northern Utah (Fig. 13c). Finally, for the NWIVT composite, the high $IVT$ values move across the Cascade Mountains and then over the low-elevation Snake River Plain to northern Utah (Fig. 13d). In each case, 700-hPa winds near 15 m s$^{-1}$ exceed the most frequent cool-season climatological wind speeds in the central Wasatch by roughly 10 m s$^{-1}$ (see Fig. 7a). Composite 700-hPa temperatures in the central Wasatch are roughly -3 °C for each synoptic type, aside from NWIVT events, where they are closer to -5 °C (Fig. 13a-d).

These are relatively warm temperatures for mountain crest level in the central Wasatch (Fig. 9), indicative of the warm nature of these high moisture transport systems, which often produce heavy LPE and modest snowfall totals. For each composite, the mean SLR is less than 9, substantially less than the mean SLR of 13.8 at CLN (Fig. 5), but comparable with typical SLR values during 95$^{th}$ percentile LPE events (Fig. 8b).

While over half of the 95$^{th}$ percentile LPE events fall into one of the high $IVT$ synoptic types, orographic storms in the central Wasatch can occur under other synoptic environments not associated with high integrated vapor transport. Northwesterly flow post-cold-frontal orographic convection storms are substantially colder than the high $IVT$ synoptic types, with mean 700-hPa temperatures near -9 °C (Fig. 14a); additionally, $IVT$ values are much lower, with mean values in the 50-150 kg m$^{-1}$ s$^{-1}$ range (Fig. 14a). This synoptic type also has lower wind speeds from the northwest that are closer to the climatological mean wind speed and direction (see Fig. 7a). As evidenced by the tight packing of 700-hPa temperature contours in Arizona and New Mexico (downstream of Utah), the environment is clearly post-frontal (Fig. 14a). Given the cold nature of NWPF storms (Figs. 9, 14a), the mean SLR for these events is 12.4, higher than the SLR values for the high $IVT$ synoptic types; this is consistent with nearly 50% of snow events being NWPF, because extreme snow events tend to have higher SLRs (Figs. 8a, 12). When we conduct the same analysis for snow events in the NWPF classification, the mean SLR is 17.7 (not shown). During frontal events, a front can stall over the Wasatch with a distinct wind shift from southwesterly to northwesterly (Shafer and Steenburgh 2008). The tight packing of the dashed red temperature contours in Fig. 14b characterizes the distinct airmass change in FR events, where intense precipitation rates and post-frontal cold air advection are common. Closed low pressure systems were the most variable in our analysis, with a wide range of locations of the low pressure center, but the mean low pressure center was located in northern Utah (Fig. 14c).

Vapor Transport Diagnostics and Precipitation Efficiency

Our analysis demonstrates that extreme orographic snowfall in the central Wasatch can be associated with high or low $IVT$ synoptic environments. Here, we present the relationships between time-integrated vapor transport ($TIVT$) and new snow and LPE in the central Wasatch. We calculate $TIVT$ as:

$$TIVT = \int_{t_i}^{t_f} IVT\, dt, \tag{2}$$

where $t_i$ and $t_f$ are the initial and final times in a 12-h observation period, respectively. Snow events have relatively low $TIVT$ compared to LPE events, with values for the former $\leq 0.6 \times 10^7$ kg m$^{-1}$ for all but 5 events (cf. Fig. 15a, b). Snow events are most frequently NWPF, which contain low $IVT$, as evidenced by the tight clustering of orange points with northwesterly crest-level flow in Fig. 15a. These snow events rarely feature high $TIVT$, although 18.1% were SWIVT events, and one had $TIVT > 0.8 \times 10^7$ kg m$^{-1}$ (Fig. 15a).

Extreme LPE events in the central Wasatch skew towards higher $IVT$ compared to extreme snow events, with about half of the former containing $TIVT \geq 0.6 \times 10^7$ kg m$^{-1}$ and the majority falling in one of the four high $IVT$ synoptic types (Fig. 15b). Some events in the high $IVT$ classifications have relatively low $TIVT$ because they are decaying atmospheric rivers with high $IVT$ upstream and substantial vapor depletion before reaching the central Wasatch. Despite typically containing high SLRs, the NWPF synoptic type was the second most frequent for extreme LPE events, suggesting that post-frontal orographic forcing or GSL-effect snow can still produce heavy LPE despite high SLRs and low $TIVT$. One NWPF event, for instance, was one of the ten largest LPE events in our analysis and had $TIVT \cong 0.2 \times 10^7$ kg m$^{-1}$ (Fig. 15b). On the contrary, six of the ten largest LPE events had $TIVT > 0.6 \times 10^7$ kg m$^{-1}$ (Fig. 15b). The large diversity of flow directions for FR and CL snow and LPE extremes is indicative of the tendency for these types of storms to approach northern Utah from many directions.

We now examine the efficiency of integrated vapor transport conversion to observed LPE. We define precipitation efficiency ($PE$) as the ratio of LPE ($P$) to the vapor-derived total precipitation ($VTP$):

$$PE = \frac{P}{VTP}, \tag{3}$$

where

$$VTP = \frac{TIVT \times 1\mathrm{m}}{1\mathrm{m}^2}. \tag{4}$$

Multiplying $TIVT$ by 1 m compresses the column integrated vapor transport down to an infinitesimally thin layer of vapor and dividing by 1 m$^2$ results in a per unit area quantity. Precipitation efficiencies for central Wasatch storms have small values on the order of $10^{-5}$, suggesting that even during extreme events most vapor transport *is not* converted to observed precipitation (Fig. 16). The median precipitation efficiency is highest when the 700-hPa flow direction is west-northwesterly or northwesterly, likely due to orographic enhancement mechanisms or the GSL-effect and a smaller moisture transport in events with those flow directions, resulting in a smaller denominator in Eq. (3) (Fig. 16). The difference in median $PE$ between WNW-NW flow and other flow directions is statistically significant at the 95$^{th}$ percent confidence interval for LPE events (Fig. 16b). The two outlier events in WNW and NW flow with $PE > 3.5 \times 10^{-5}$ occurred during a period of lake-effect precipitation in October 2010. Lake-effect snowfall likely has a high precipitation efficiency due to the limited vapor transport during these events, with fairly cool air flowing over the GSL triggering impressive snow and LPE amounts, despite there being minimal vapor transport in the atmosphere.

Relationship Between High $IVT$ and Observed Precipitation

Our results and prior research (e.g., Rutz et al. 2014; their Fig. 2a) imply that there is no obvious link between $IVT$ and observed LPE or snow in the central Wasatch. Events with high *or* low $TIVT$ can produce the largest 12-h LPE values at CLN during a 23-year period (Fig. 15b). In this section, we examine the characteristics of high $IVT$ events, defined as events with $IVT \geq 200$ kg m$^{-1}$ s$^{-1}$ (the 98.9$^{th}$ percentile of $IVT$ over the central Wasatch) at the mid-point of a 12-h observing period at CLN. There were 112 of such periods, and 37 produced *no* LPE at CLN, while 19 resulted in a 95$^{th}$ percentile LPE event at CLN ($\geq 27.9$ cm in 12-h). High $IVT$ periods that produced heavy LPE were colder than those that did not produce LPE, with median temperatures ~10 °C colder throughout the atmosphere for the former (Fig. 17a). Additionally, extreme-LPE-producing events have higher wind speeds and winds that feature more westerly to northwesterly flow than events that do not produce LPE (Fig. 17b, c). This is consistent with westerly to northwesterly winds at 700-hPa having higher precipitation efficiency than other flow directions (Fig. 16). A major difference between high $IVT$ events that do not produce LPE at CLN and ones that do is that the atmosphere is near saturated at all levels for extreme-LPE-producing events, while relative humidity values are much lower for ones that do not generate precipitation (Fig. 17d). As expected, given the lower wind speeds for events which do not produce LPE (Fig. 17b), specific humidity is higher for events which do not produce LPE (Fig. 17e). Lastly, vertical velocity values are much higher for high $IVT$ periods that which produce extreme LPE than for those that do not, especially near and above mountain crest-level, likely due to Rossby waves and frontogenetical dynamics during extreme LPE periods (Fig. 17f). In this analysis, we find that that high $IVT$ *can* be an orographic precipitation-generating mechanism in the central Wasatch, but it does not necessarily guarantee extreme LPE. For periods with elevated $IVT$, additional environmental factors are needed to produce extreme LPE, including low temperatures, westerly to northwesterly flow, a saturated atmosphere, and high upward vertical velocities.

Radar Characteristics

We conclude our analysis of the characteristics of extreme orographic snowfall in the Wasatch by presenting level II statistics from the KMTX radar. We produced plots of the frequency of 0.5° radar echoes ≥ 10 dBZ for both snow and LPE extremes, binned by either flow direction or synoptic classification. The structure of these plots appeared the same for both snow and LPE extremes, so for brevity, we chose to only present radar data for LPE extremes. Fig. 18 displays the frequency of 0.5° radar echoes ≥ 10 dBZ for SSW, SW, WSW, W, WNW, and NW winds. During periods with southwesterly and west-southwesterly flow, there is broad coverage of radar echoes in the mountains *and* adjacent valleys (Fig. 18a-c), whereas in west-northwesterly to northwesterly flow, radar echoes are confined to the central Wasatch (Fig. 18d-e), likely due to post-frontal orographic or lake-effect convection. In contrast to prior radar analysis in other regions which finds that orographic precipitation most frequently occurs if the flow is orthogonal to the terrain (e.g., Houze et al. 2001; Yuter et al. 2011), the flow direction has no distinct impact on the distribution of orographic precipitation *within* the central Wasatch mountains. It does, however, impact the distribution of lowland precipitation in the upstream valleys.

Our radar analysis of synoptic types yielded similar results to the analysis for wind directions. Storms associated with high $IVT$ tend to have a broad distribution of radar echoes, especially when the flow is from the southwest or west (Fig. 19a-d). The confinement of radar echoes to the mountains during high $IVT$ southerly and northwesterly flow systems could be due to the small sample sizes for each of those events. Similar to the cases of NW and WNW flow (Fig. 18e-f), NWPF events feature radar echoes that are primarily confined to the central Wasatch (Fig. 19e). Frontal events have the highest percentage of echoes ≥ 10 dBZ across the mountains and valley, suggesting that these events tend to reduce the ratio of mountain to valley precipitation (Fig. 19f). Although they have a small sample size, closed low pressure systems tend to feature a broad coverage of radar echoes in both mountain and valley locations (Fig. 19g), similar to prior findings on the distribution and intensity of precipitation during closed lows in Northern Utah (e.g., Williams and Peck 1962).

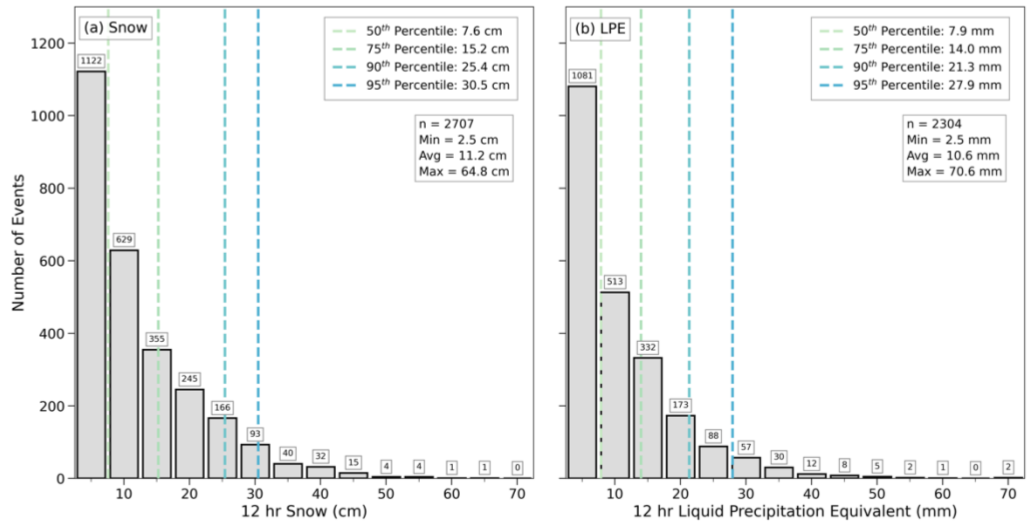


Figure 2. Histogram of 12-hr snow events at CLN during the 2000–2022 cool seasons. Bins are in 5-cm increments starting at 2.5 cm. The dashed vertical lines indicate the 50th, 75th, 90th, and 95th percentiles. Numbers on top of each bar are the number of events in that bin. (a) Snow events. (b) LPE events.

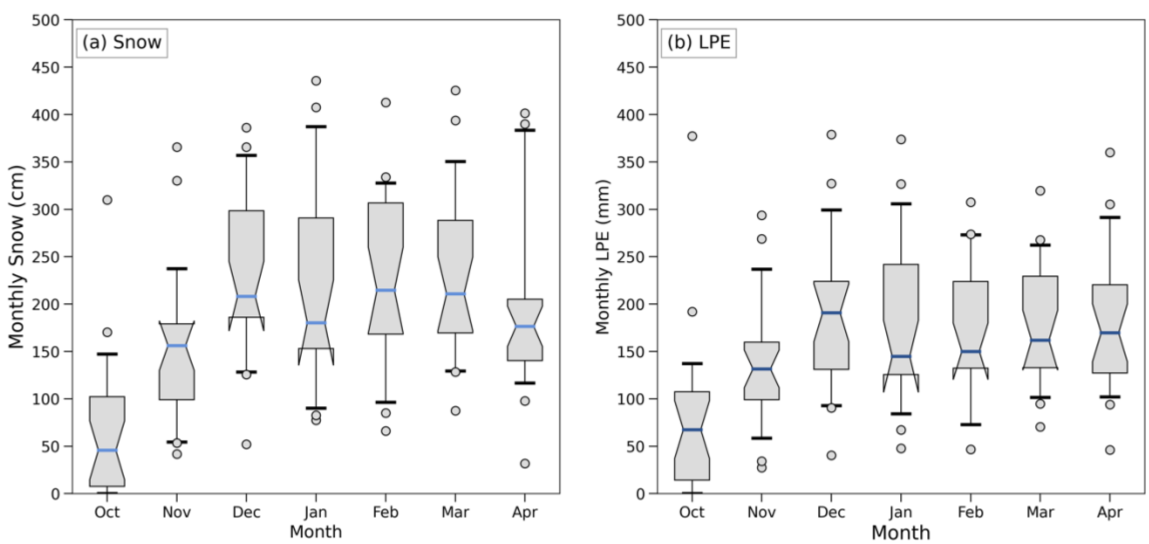


Figure 3. Box-and-whisker plots of monthly (a) snow and (b) LPE at CLN during the 2000–2022 cool seasons. The colored horizontal line in each box represents the median accumulation, the box extends from the 25th to 75th percentile, the whiskers are the 5th and 95th percentile values, and the circles indicate outlier values. Notches indicate the 95th percent confidence interval of the median.

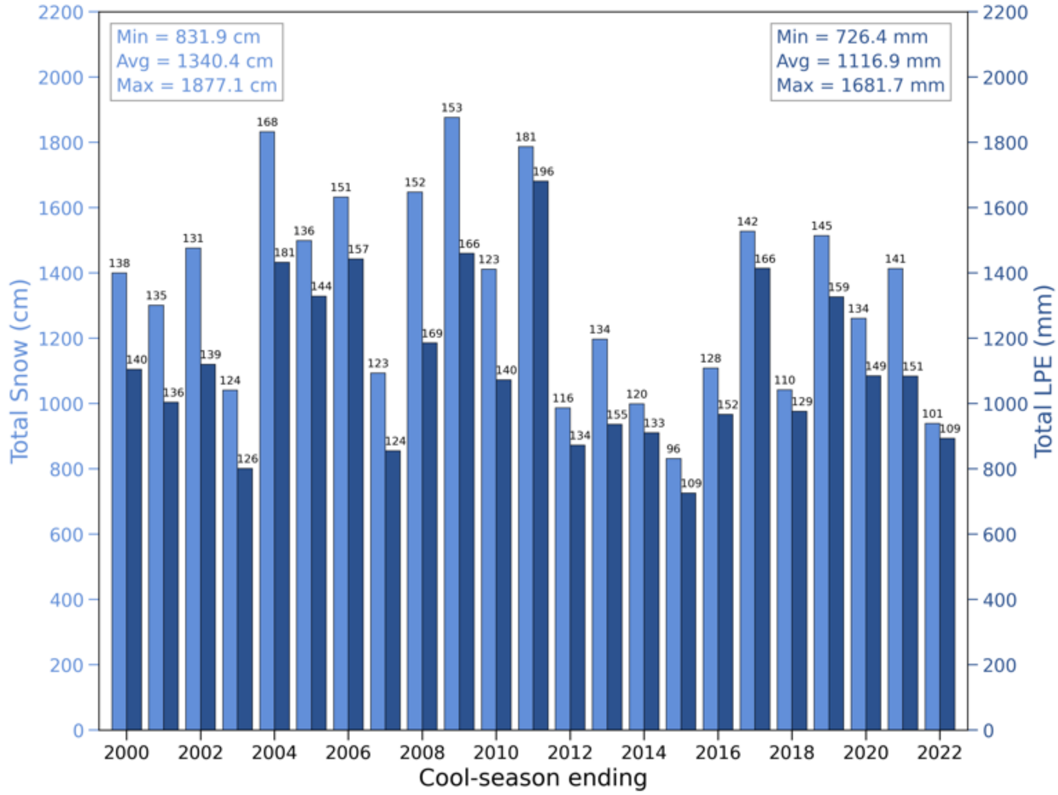


Figure 4. Total snow (light blue) and LPE (dark blue) during the 2000–2022 cool seasons. Numbers on top of each bar indicate the number of events that occurred during that season.

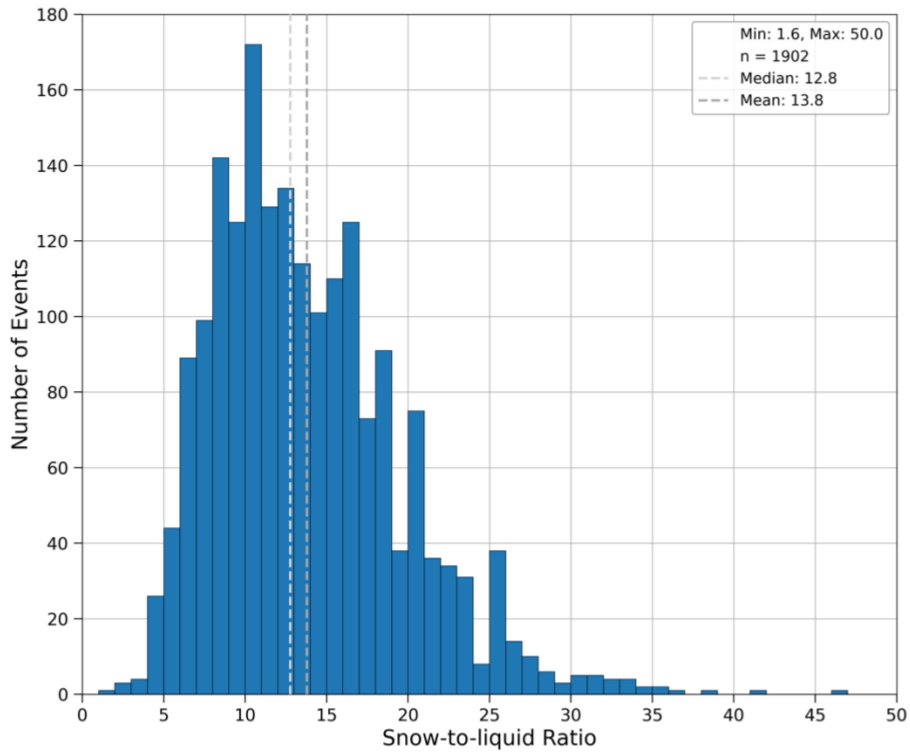


Figure 5. Histogram of snow-to-liquid ratio for all events at CLN with ≥ 5.1 cm of snow and ≥ 2.8 mm of LPE. Bins have a width of 1. The light and dark grey lines indicate the mean and median SLR, respectively.

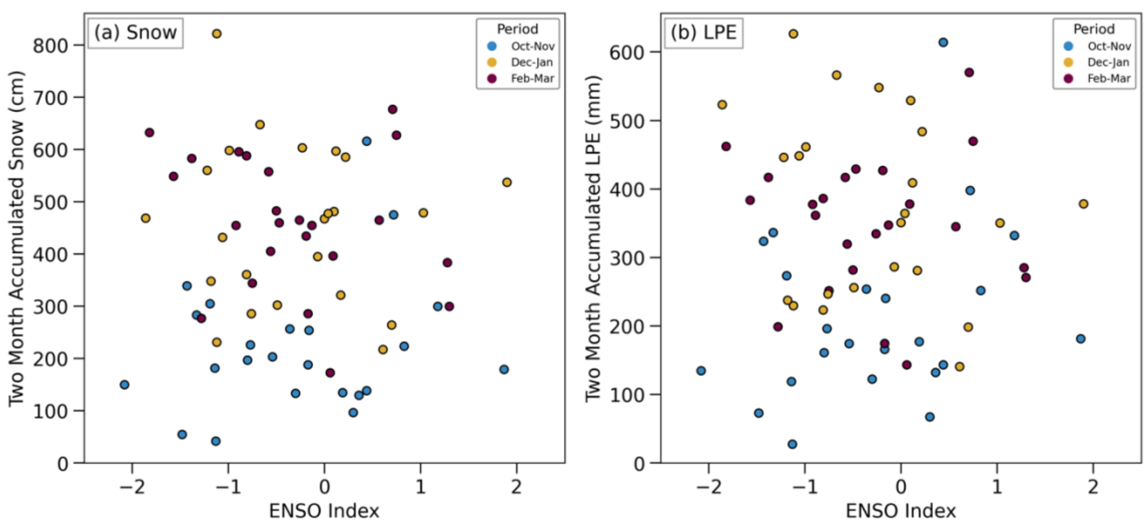


Figure 6. Scatter plot of ENSO index and (a) two-month accumulated snow and (b) two-month accumulated LPE at CLN.

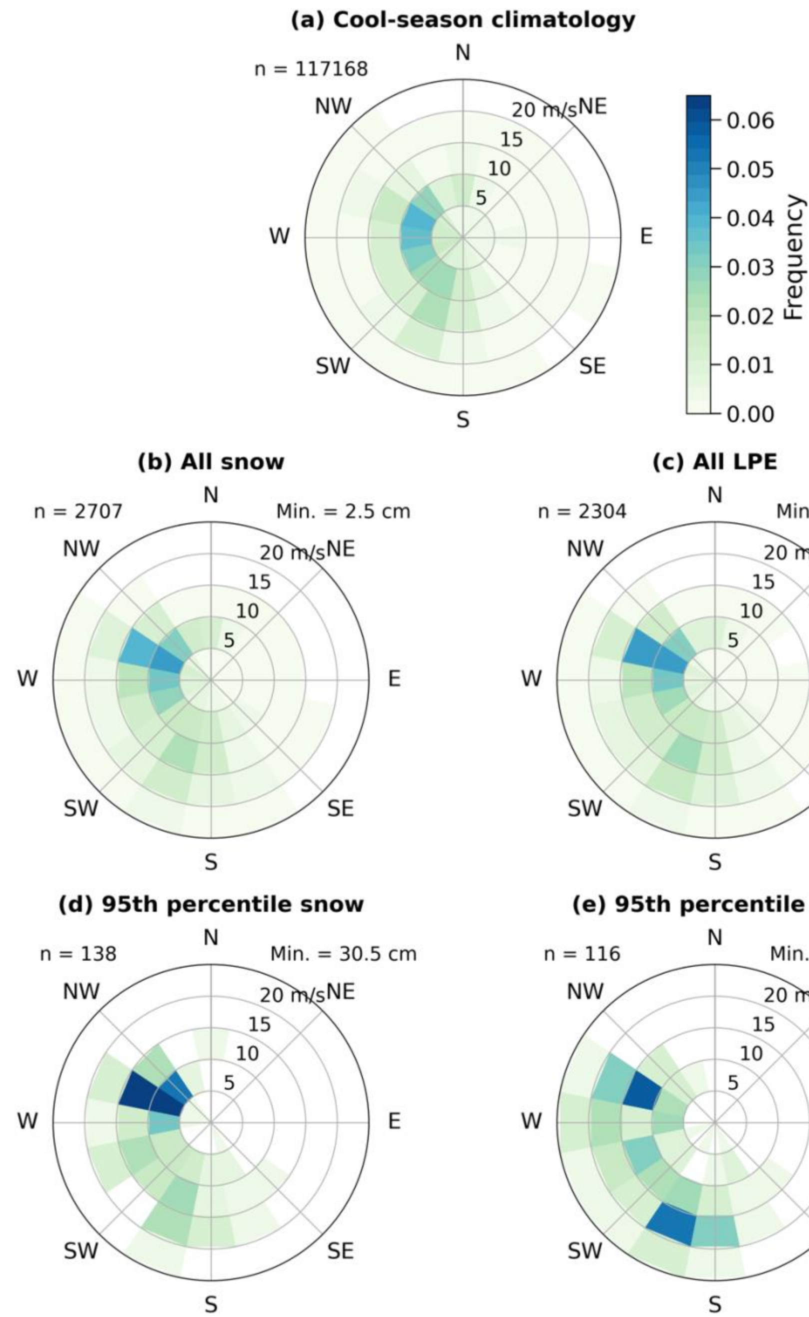


Figure 7. Normalized 2-D histogram of the frequency of 700-hPa wind speed and direction in the central Wasatch. Bins are in 5 m $s^{-1}$ and 22.5° increments. (a) Cool-season climatology. (b-e) As in (a) but for all snow events, all LPE events, 95$^{th}$ percentile snow events, and 95$^{th}$ percentile LPE events, respectively.

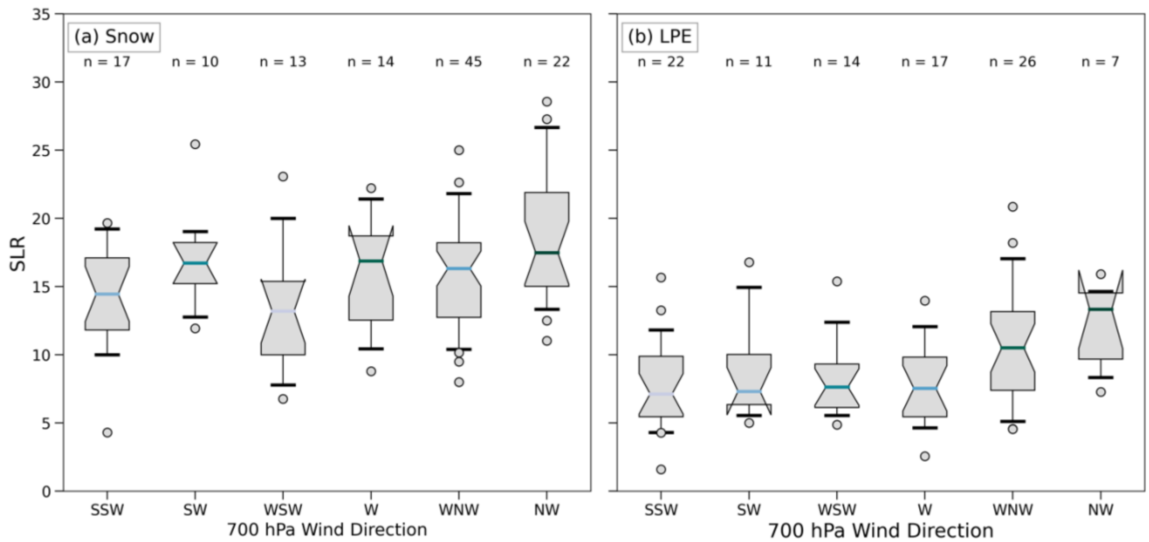


Figure 8. Box-and-whisker plots of SLR for 95$^{th}$ percentile events binned by 700-hPa wind direction. The colored horizontal line in each box represents the median SLR, the box extends from the 25$^{th}$ to 75$^{th}$ percentile SLR, the whiskers are the 5$^{th}$ and 95$^{th}$ percentile values, and the circles indicate outlier values. Notches indicate the 95$^{th}$ percent confidence interval of the median. (a) Snow events. (b) LPE events.

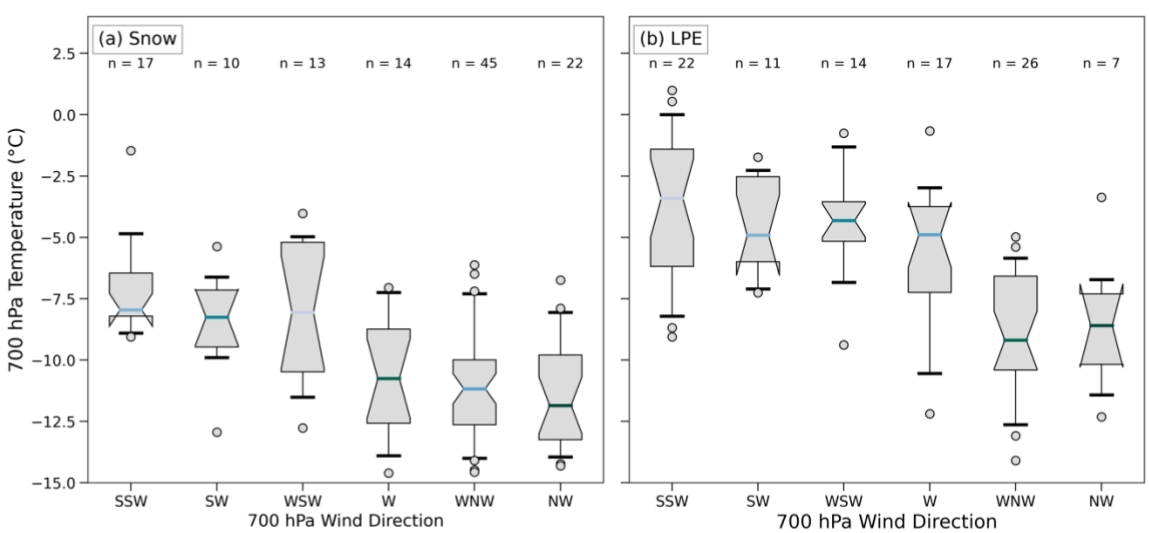


Figure 9. As in Fig. 8, but for 700-hPa temperature.

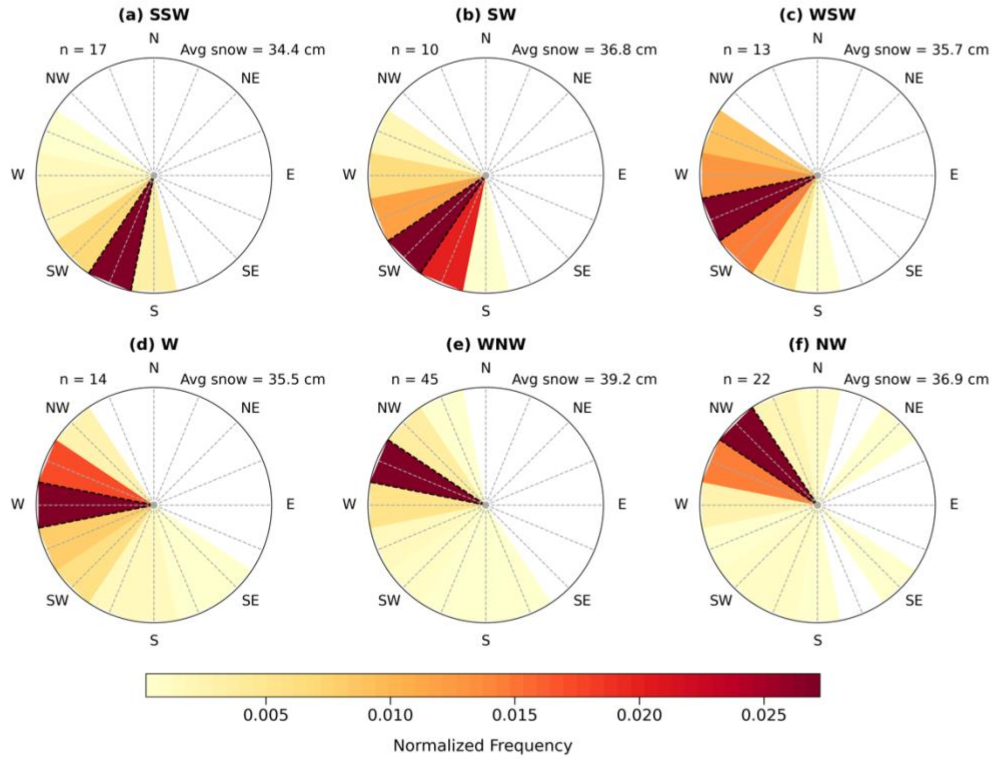


Figure 10. Frequency of 700-hPa wind direction during each 95th percentile snow event binned by wind direction classification (color bar at bottom). Number of events is listed in the upper left corner. (a) South-southwesterly flow. (b-g) As in (a), but for southwesterly, west-southwesterly, westerly west-northwesterly, and northwesterly flow.

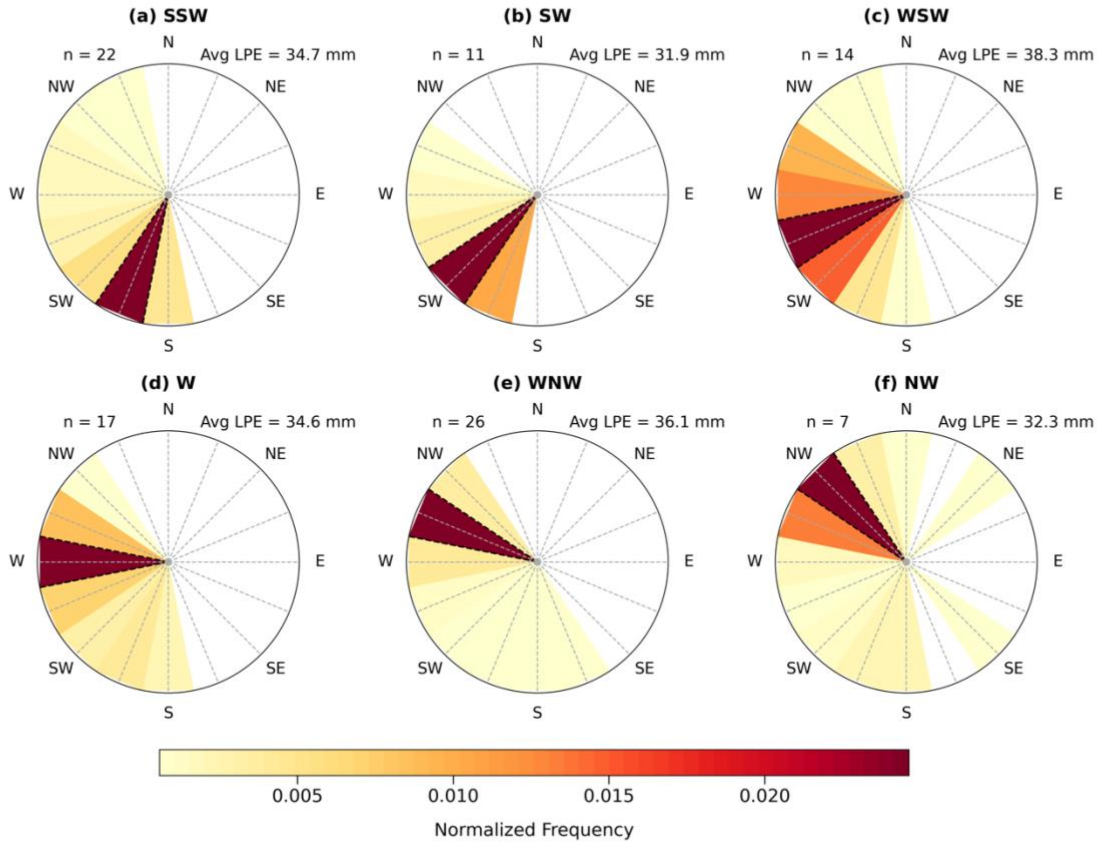


Figure 11. As in Fig. 10, but for 95$^{th}$ percentile LPE events.

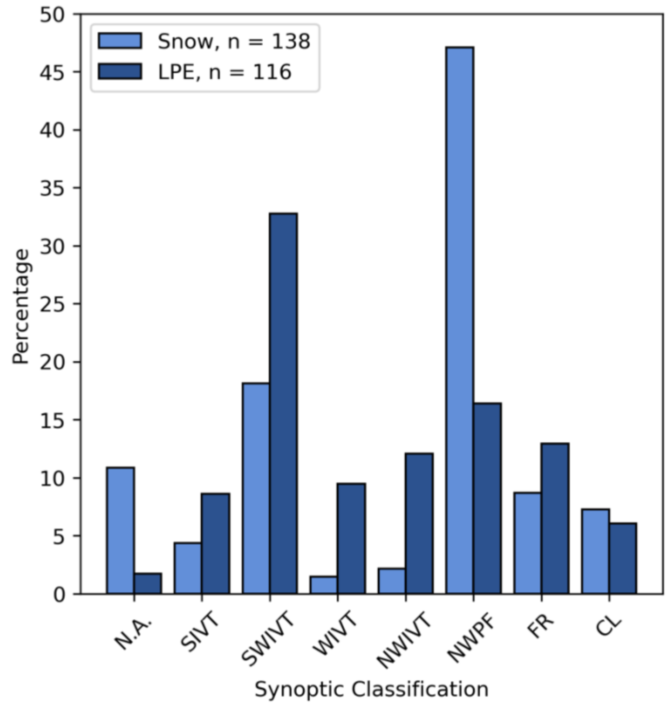


Figure 12. Percentage of 95th percentile snow and LPE events by synoptic classification.

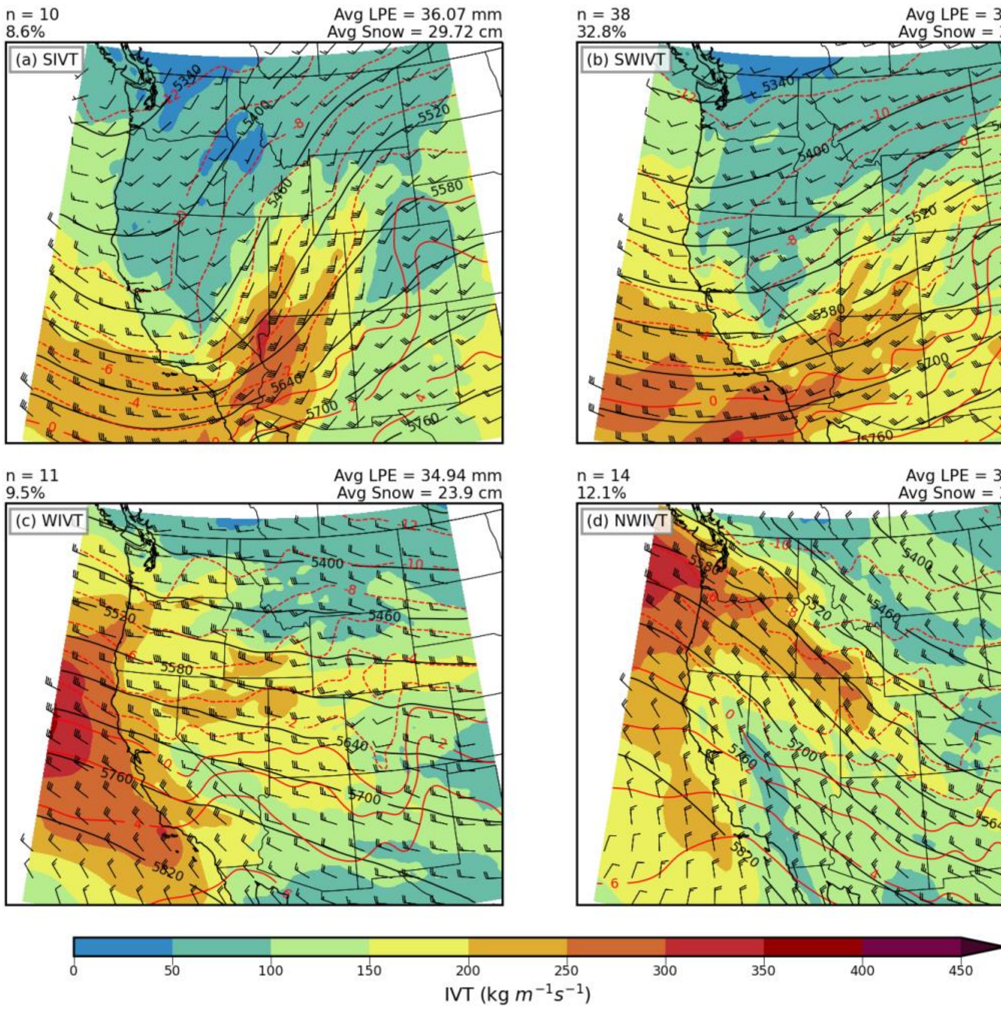


Figure 13. Composites of high $IVT$ synoptic classifications for 95th percentile LPE events. (a) $IVT$ (kg m$^{-1}$ s$^{-1}$; color-fill scale at bottom), 500-hPa geopotential height (m; black contours), 700-hPa temperature (°C; red contours), and 700-hPa wind (barbs; full barb = 10 kt) for SIVT events. The number of events and percent of 95th percentile LPE events in the SIVT classification annotated at upper left and the mean LPE and snow for 95th percentile LPE events annotated in the upper right corner. (b-d) As in (a), but for events in the SWIVT, WIVT, and NWIVT classifications.

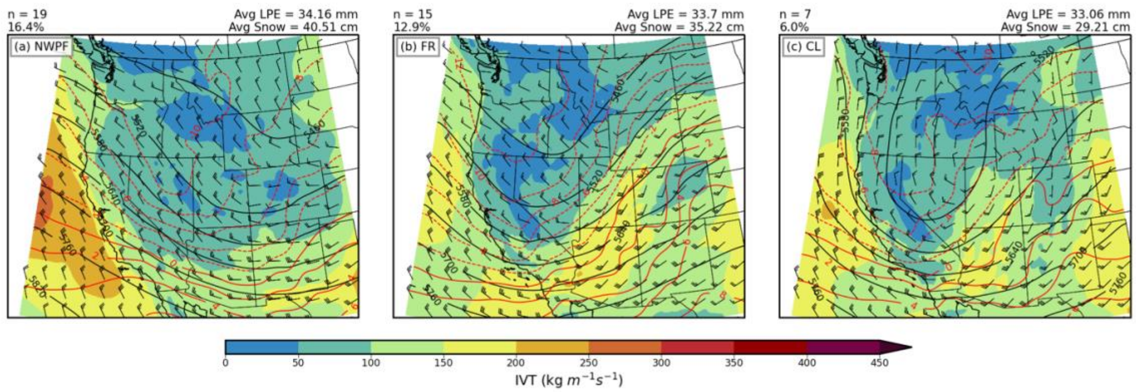


Figure 14. As in Fig. 13, but for 95th percentile LPE events in the (a) NWPF, (b) FR, and (C) CL synoptic classifications.

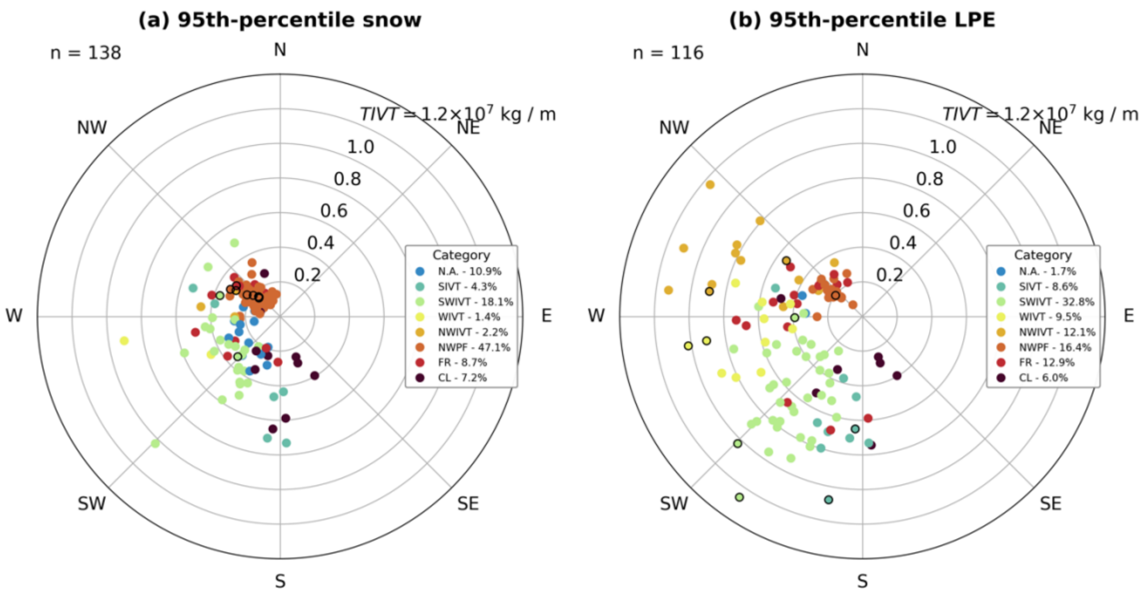


Figure 15. Polar plot of time integrated vapor transport and 700-hPa wind direction for all 95th percentile events colored by synoptic classification. The ten largest events are outlined in black. (a) Snow. (b) LPE.

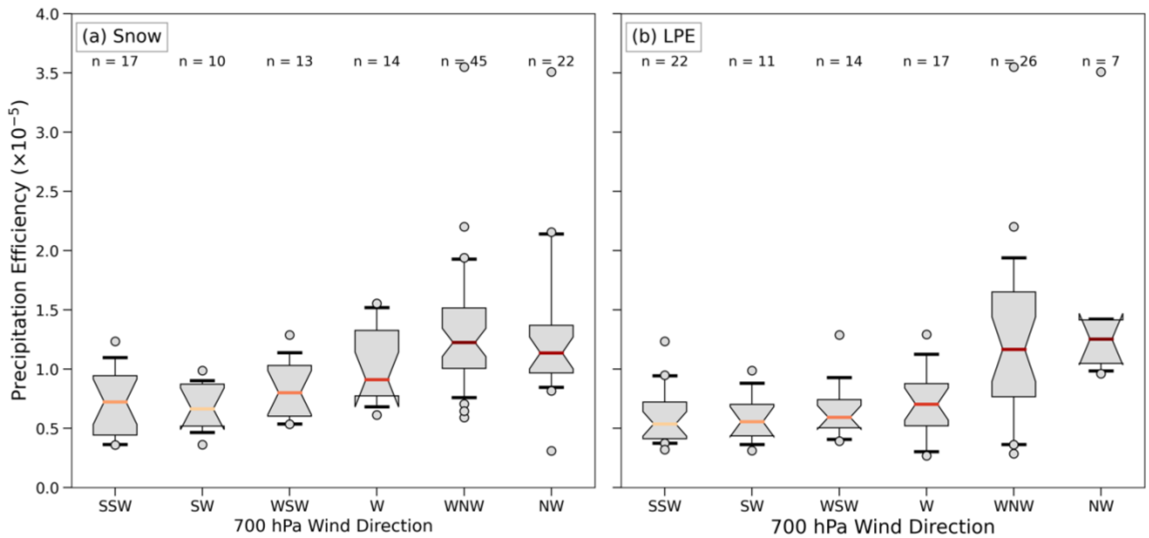


Figure 16. As in Fig. 8 but for precipitation efficiency.

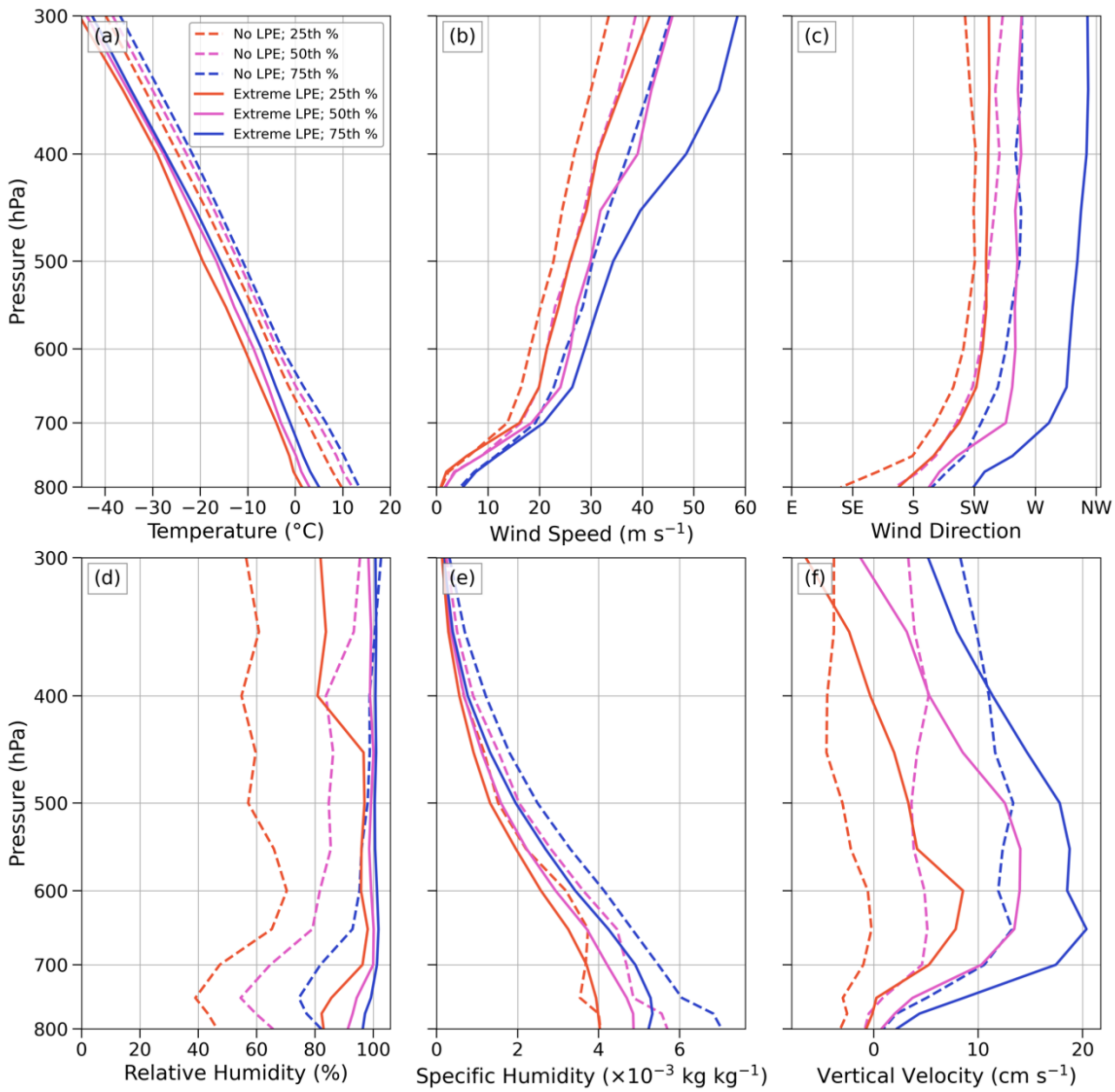


Figure 17. Atmospheric profiles for periods with $IVT \geq 200$ kg m$^{-1}$ s$^{-1}$ with no LPE recorded at CLN (dashed lines) and 95$^{th}$ percentile LPE at CLN (solid lines). Colors indicate profile percentiles [see legend in (a)]. (a) Temperature. (b) Wind speed. (c) Wind direction. (d) Relative humidity. (e) Specific humidity. (f) Vertical velocity.

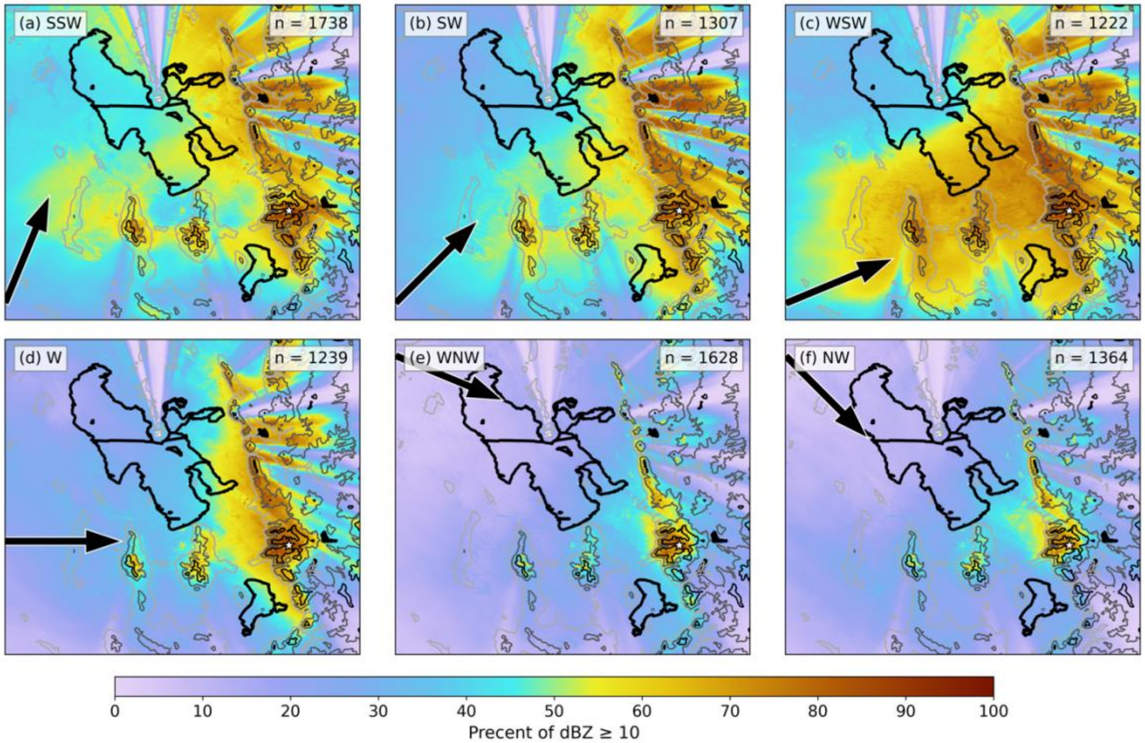


Figure 18. Frequency of 0.5° radar echoes ≥ 10 dBZ (%; scale at bottom) for radar scans occurring during 95th percentile LPE events with south-southwesterly 700-hPa winds. Thick black lines outline lakes and thin contours display terrain height from 1200 m MSL to 2700 m MSL, every 500 m. Black arrow displays the prevailing 700-hPa flow direction and the white star denotes CLN. The number of radar scans used to produce the plot is in the upper right corner. (a) South-southwesterly flow. (b-f) As in (a), but for events southwesterly, west-southwesterly, westerly, west-northwesterly, and northwesterly flow.

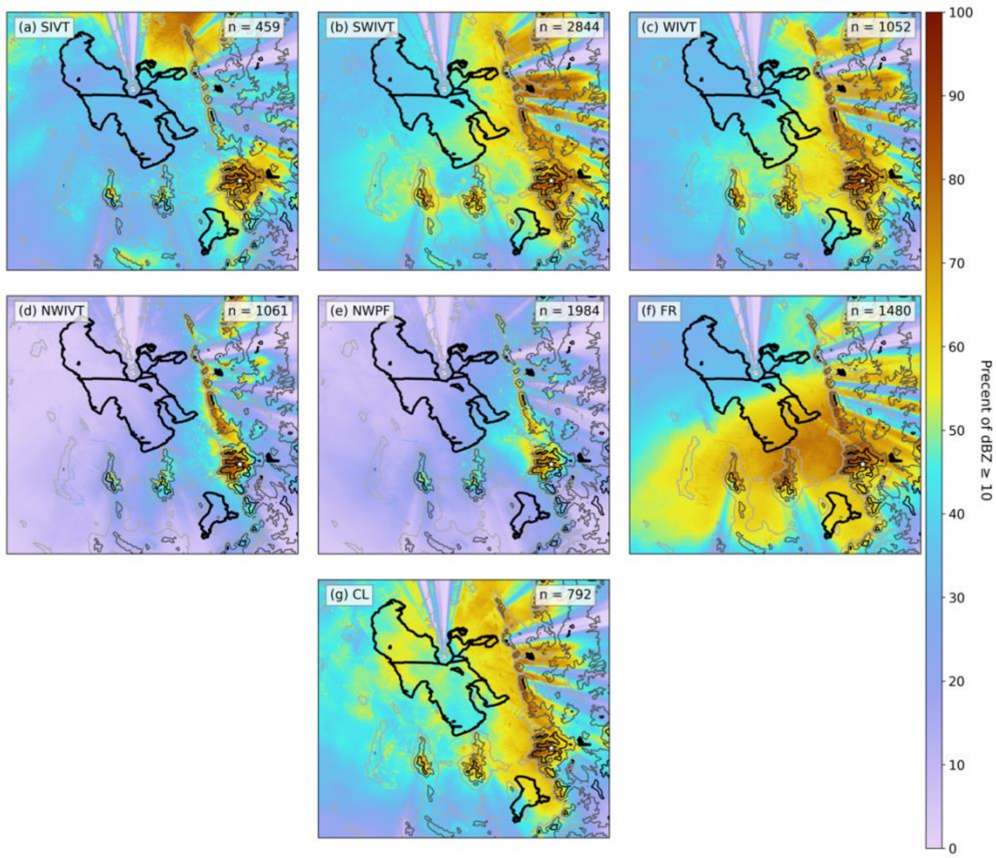


Figure 19. As in Fig. 18 but for synoptic classifications of 95th percentile LPE events for (a-g) SIVT, SWIVT, WIVT, NWIVT, NWPF, FR, and CL.

# 4. CONCLUSIONS

Using snow and LPE observations from the CLN snow study plot at Alta Ski Area, this research examined the characteristics of extreme orographic snowfall in northern Utah's upper Little Cottonwood Canyon, one of the snowiest and most avalanche-prone places in the interior western United States. During the 2000–2022 cool-season (Oct–Apr) study period, there was an average of 117.7 12-h periods with $\geq 2.54$ cm of new snowfall at CLN, the largest of which produced nearly 65 cm of snow. Almost all precipitation at CLN falls as snow during the cool season, with median snow-to-liquid ratios of 13.3, characteristic of an intermountain snow climate. Extreme events, defined as the 95$^{th}$ percentile of new snow or LPE, exceeded 30.5 cm and 27.9 mm, respectively. Northern Utah lies at the nodal point in precipitation tendencies resulting from the El Niño Southern Oscillation, and data from CLN confirm this, as there is no correlation between ENSO phase and snow or LPE at CLN.

There is a large diversity of storm tracks and types which produce extreme orographic snowfall in the central Wasatch, resulting in highly variable snowfall characteristics. Although 95$^{th}$ percentile snow events most frequently occur in west-northwesterly and northwesterly flow and tend to contain lower temperatures and SLRs, 95$^{th}$ percentile LPE events are most common during south-southwesterly flow, with higher temperatures and SLRs. Despite these frequency maxima being present in our analysis, extreme snow and LPE at CLN *can* occur across a wide range of crest-level flow directions. This is in contrast to other mountainous regions, where extreme precipitation is highly dependent on wind direction (e.g., Yuter et al. 2011). Most events at CLN see high flow persistence, with little change in the 700-hPa wind direction over the course of an event. However, there is a tendency towards veering winds during an event. While most snow events are associated with northwesterly flow post-cold-frontal orographic convection, LPE events are more common during periods of high integrated vapor transport from a wide range of flow directions, demonstrating the high variability of storm types associated with extreme orographic snow in the Wasatch.

While other mountainous regions throughout the world have a high correlation between integrated vapor transport and LPE (e.g., Dettinger et al. 2011; Ricciotti and Cordeira 2022), the relationship between $IVT$ and LPE and snow in the Wasatch is not as direct. Storms with relatively low $IVT$ can still produce heavy LPE, often in northwesterly flow due to orographic forcing mechanisms or the GSL-effect — these storms have high precipitation efficiency. On the contrary, storms with high $IVT$ can also produce significant LPE in the Wasatch, but less of the vapor fluxed over the region is converted into observed precipitation. Furthermore, we found that for periods with $IVT \geq$ kg $m^{-1}$ $s^{-1}$, atmospheric conditions have a strong impact on whether there is no LPE or an orographic snowfall extreme. These findings imply that observed LPE and snow in interior mountain ranges have a substantially smaller correlation with moisture transport than coastal mountain ranges.

Our radar analysis reveals the high variability in the spatial distribution of precipitation near and in the central Wasatch in storms with different flow directions and synoptic types. In storm environments with flow directions ranging from south-

southwesterly to westerly, radar echoes are broadly distributed in mountains *and* upstream valleys, while northwesterly flow post-cold-frontal environments feature echo confinement in the central Wasatch. This contrasts with prior research in other mountain ranges, where the greatest frequency of radar echoes is often located on or upstream of ridgelines oriented orthogonally to the flow (e.g., Houze et al. 2001; Yuter et al. 2011). The ability for many flow directions to produce extreme orographic snowfall in the Wasatch is indicative of the range's complex and varied topography, which can be orthogonal to the flow in several flow regimes. Synoptic types associated with high $IVT$, frontal systems, and closed low pressure systems produce precipitation in the central Wasatch and adjacent valleys, while post-frontal northwesterly flow features orographic enhancement that limits observed precipitation to the central Wasatch.

Although this research focused on the central Wasatch range, it has broader implications for understanding orographic precipitation in mountain ranges throughout the world. As a mid-latitude range, the Wasatch can see snowfall in a variety of cool-season storm tracks and flow directions, and even during split-flow periods. While the Wasatch is a narrow range, the terrain of the *central Wasatch* broadens and has a nearly 1:1 horizontal aspect ratio (see Fig. 1), enabling orographic precipitation in a large variety of flow directions. We hypothesize that interior mountain ranges are most optimally suited to receive heavy cool-season orographic precipitation when they fall at the center of the predominate storm track and feature terrain with a roughly horizontal 1:1 horizontal aspect ratio.

Our results in this paper extend beyond prior research by 1) comprehensively analyzing orographic storms in the central Wasatch, 2) examining meteorology associated with extreme orographic precipitation from a snow and LPE perspective, and 3) drawing linkages between $IVT$ and observed precipitation in an *interior* mountain range. In addition to improving our understanding of cool-season orographic snowfall extremes in the central Wasatch, this research will develop further knowledge of orographic precipitation processes in interior mountain ranges. Future research could analyze the characteristics of individual extreme storms in the central Wasatch through real-data modeling to enhance knowledge gained in this study.

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